Module - I

SETS, FUNCTIONS AND RELATIONS

Syllubus

Sets and subsets, Venn Diagrams, Set operations, set Identities, Generalized union and intersection. The principle of inclusion - Exclusion (Basic and generalized version) and applications.

Function definition, injections, surjections and Bijections, Inverse functions, and Compositions of functions, Cardinality of sets, Cantor diagonalization argument.

Relations and Their properties, composition of relations, n-ary relations. Representing relations using matrices

Equivalence relations, Equivalence classes, Partial ordering, Hasse Diagrams, Maximal and minimal elements, Lattices.

Sets

A set may be viewed as an unordered collection of distinct objects, called as elements or members of the set. We ordinarily use capital letters A, B, X, Y, ... to denote sets, and lower case letters a, b, x, y, ... to denote elements of sets.

- Let A be a set, then 'p is an element of A' or 'p belongs to A' is written as

$$p \in A.$$

- 'p not an element of A' is written as

$$p \notin A.$$

Principle of Extension

Two sets A and B are equal if and only if they have the same members.

If the sets A and B are equal, we write $A = B$. Otherwise $A \neq B$.

Specifying set

There are essentially two ways to specify a set. One way, if possible, is to list its members. known as roster form.

Eg: $A = \{a, e, i, o, u\}$

Another way is set builder form. Which is to state those properties which characterized the elements in the set.

Eg: $B = \{x : x \text{ is an even integer}, x > 0\}$.

Some examples for sets

N = the set of positive integers : 1, 2, 3, . . .

Z = the set of integers : −2, −1, 0, 1, 2, . . .

Q = the set of rationals

R = the set of real numbers

C = the set of Complex numbers.

Universal set

The members of all sets under investigation usually belong to Some fixed large set called the Universal set, which is denoted by U.

Eg: In plane geometry, the universal set consists of all the points in the plane.

Empty set

The set with no elements is called the empty set or null set, and is denoted by ϕ

Eg: Consider

$$S = \{x : x \text{ is a positive integer}, x^2 = 3\} = \phi.$$

Subsets

If every element in a set A is also an element of a set B, then A is called a subset of B, we also say that A is contained in B or that B contains A. This relationship is written

$$A \subseteq B \text{ or } B \supseteq A$$

If A is not a subset of B ie, If atleast one element of A does not belong to B, we write $A \nsubseteq B$ or $B \nsupseteq A$

If $A \subseteq B$, then it is still possible that $A = B$, When $A \subseteq B$ but $A \neq B$, we say A is a proper subset of B. we write $A \subset B$.

Examples: 1. $A = \{1,3\}$, $B = \{1,2,3\}$, $C = \{1,3,2\}$.

A and B are subsets of C but A is a proper subset of B and C.

but B is not a proper subset of C.

2. Consider N, Z, Q, R, C, then

$N \subseteq Z \subseteq Q \subseteq R \subseteq C$ or $N \subset Z \subset Q \subset R \subset C$

Properties

i $A \subseteq U$, $\phi \subseteq A$.

ii $A \subseteq A$.

iii $A \subseteq B$, $B \subseteq C \Rightarrow A \subseteq C$.

iv $A \subseteq B$ and $B \subseteq A \Rightarrow A = B$.

Conversely if $A = B \Leftarrow A \subseteq B, B \subseteq A$.

3. Let $A = \{a\}$, $B = \{a, \{b, c\}, d\}$, $C = \{a, \{b,c\}, d, \{e\}\}$.

$A \subseteq B$, $B \subseteq C$, $A \subseteq C$.

4. $\{3\}$ Consider $A = \{1,3,5\}$.

$\{3\} \in \{1,3,5\} \Rightarrow$ False.

element $3 \in \{1,3,5\}$, but $\{3\} \notin \{1,3,5\}$.

$\{3\} \subset \{1, 3, 5\}$. True.

5. i $\phi \in \phi$ → false.

∵ An empty set does not contains any elements.

ii $\phi \subseteq \phi$ = True.

* Singlton set : A set containing only one element :

Eg : $S = \{x : x$ is a positive integer, $\sqrt{x} = 1\}$. $= \{1\}$.

* Finite set : A set having finite no: of elements.

Eg: $S = \{x : x \in N, x < 10\}$. $= \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$.

* Infinite set : A set having infinite no: of elements.

Eg: R, C, . . .

1. A set is said to be contably infinite if there is one-one correspondence between the elements in the set and elements in N. A countably infinite set is also termed as denumerable.

A set that is either finite or denumerable is called contable.

Eg: N, Z, Q, . . .

A set which is ~~con~~ not countable is called uncountable

Eg: R, C, R\Q. etc.

Power sets

For a given set S, the class of all subsets of S is called the power set of S. and will be denoted by power(s).

the no: of elements ~~of~~ in power(s) is

$$n(\text{power}(s)) = 2^{n(s)}.$$

* Cardinality of a finite set A denoted by $n(A)$ or $|A|$ is the no: of distinct elements in that set.

Eg: $A = \{1,2,3,4,5\}$.

$n(A) = 5$.

2. $B = \{1,1,4,5,6,6,7,8,9\}$

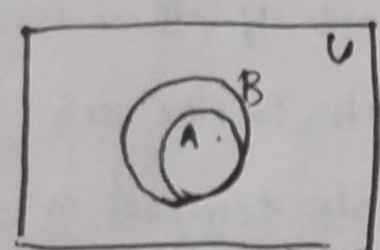
$n(B) = 7$.

Venn Diagrams

A venn diagram is a pictorial representation of sets in which sets are represented by enclosed areas in the plane.

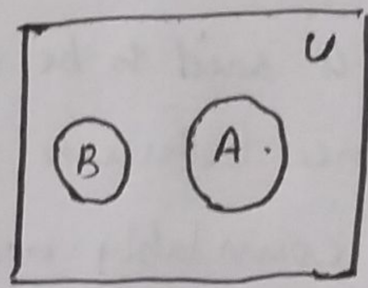
The universal set U is represented by the interior of the rectangle, and the other sets are represented by disks lying within the rectangle.

Eg: a)



$A \subseteq B$.

b)



A and B are disjoint.

* An argument is a collection of premises intended to support or infer a conclusion. Each premise must be either true or false.

Problem:

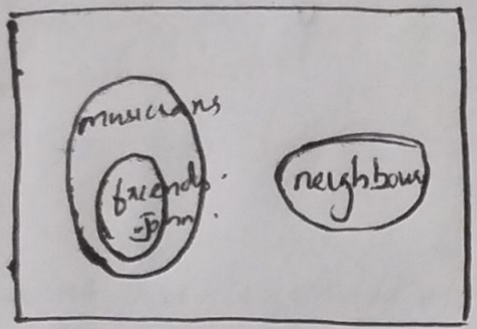
1. Show that the following argument is valid:

S_1 : ~~My Saucepans are the only~~ All my friends are musicians.

S_2 : John is my ~~fried~~ friend.

S_3 : None of my neighbours are musicians

Conclusion S: John is not my neighbour.



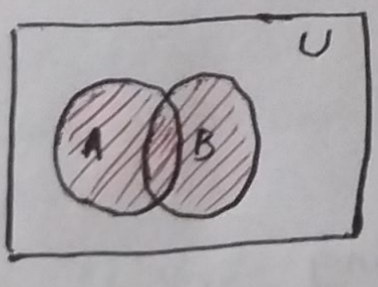
Set operations

Union of sets

The union of two sets A and B, denoted by $A \cup B$ is the set of all elements which belongs to A or to B. ie

$$A \cup B = \{x : x \in A \text{ or } x \in B\}$$

Venn diagram:

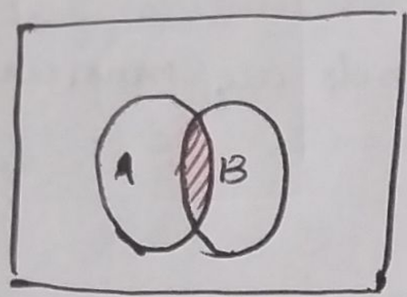


Intersection of sets

The intersection of two sets A and B, denoted by $A \cap B$, is the set of all elements which belongs to both A and B, ie

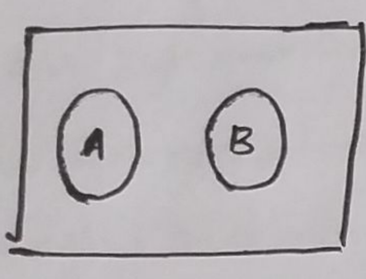
$$A \cap B = \{x : x \in A \text{ and } x \in B\}$$

Venn diagram



* If A and B do not have any elements in common, then A and B are said to be disjoint or non intersecting. ($A \cap B = \phi$)

Venn diagram.



Example:

1. Let $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6, 7\}$ $C = \{2, 3, 5, 7\}$.

Then

$A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$.

$A \cup C = \{1, 2, 3, 4, 5, 7\}$.

$A \cap B = \{3, 4\}$.

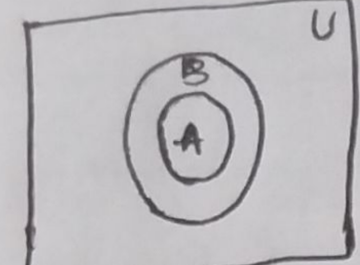
$A \cap C = \{2, 3\}$

2. Draw the venn diagram for the following

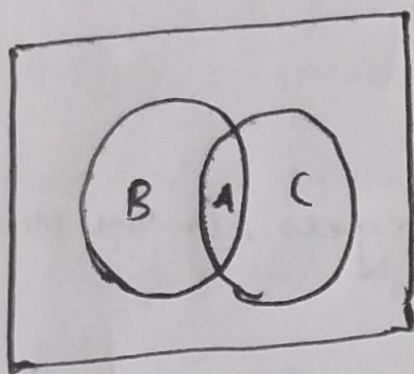
i) $(A \cap B) \subseteq B$ and $B \nsubseteq A$.

ii) $A \subseteq B$, $A \subseteq C$, $B \cap C \subseteq A$ and $A \subseteq (B \cap C)$.

i)



ii



$\because$ $B \cap C \subseteq A$ and $A \subseteq B \cap C \Rightarrow A = B \cap C$.

Theorem:

The following are equivallent:

i $A \subseteq B$

ii $A \cap B = A$.

iii $A \cup B = B$

Complements

The absolute compliment or simply compliment of a set A, denoted by A^c, is the set of elements which belong to U but which do not belong to A. ie

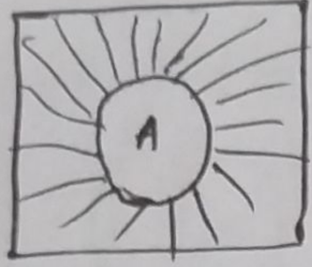
$$A^c = \{x : x \in U,\ x \notin A\}.$$

Which is also denoted by A' or $\bar{A}$.

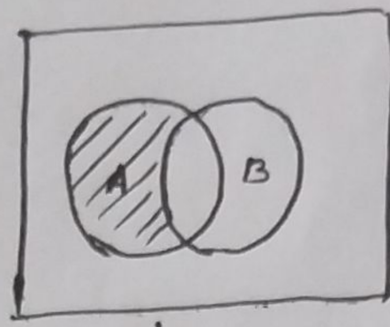
* The relative compliment of a set B with respect to a set A or simply the difference of A and B, denoted by $A \backslash B$, is the set of elements which belongs to A but which do not belong to B, ie

$$A \backslash B = \{x : x \in A,\ x \notin B\}$$

also denoted as $A - B$, $A \sim B$. or $A \cap B^c$



A^C.



$A \setminus B$.

Example:

1. Let $U = \mathbb{N} = \{1, 2, 3, \ldots\}$ the positive integers, is the universal set.

Let $A = \{1,2,3,4\}$ $B = \{3,4,5,6,7\}$ $C = \{6,7,8,9\}$

and let $E = \{2,4,6,\ldots\}$ the even integers. Then

$A^C = \{5,6,7,8,\ldots\}$ $B^C = \{1,2,8,9,10,\ldots\}$

$C^C = \{1,2,3,4,5,10,11,\ldots\}$.

And $A \setminus B = \{1,2\}$ $B \setminus C = \{3,4,5\}$ $B \setminus A = \{5,6,7\}$.

$C \setminus E = \{7,9\}$

Also $E^C = \{1,3,5,\ldots\}$ set of odd integers.

2. Find the power set of $S = \{1,2,3\}$.

Ans: power$(S) = \{\phi, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}\}$.

3. Let $U = \{1,2,3,4,\ldots 9\}$, $A = \{1,2,3,4,5\}$

$B = \{4,5,6,7\}$ $C = \{5,6,\cancel{7},8,9\}$.

Find A^c B^c, $A \cap B$, $A \cup B$, C^c, $A \cup B \cup C$, $A \cap C$.

$A \setminus B$, $B \setminus A$.

Symmetric difference

The symmetric differenie of sets A and B, denoted by $A \oplus B$, consist of those elements which belong to A or B but not to both, ie

$$A \oplus B = (A \cup B) \setminus (A \cap B).$$

One can also show that

$$A \oplus B = (A \setminus B) \cup (B \setminus A).$$

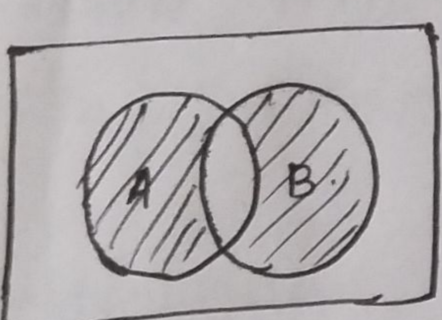
Eg:

$A = \{1, 2, 3, 4, 5, 6\}$ $B = \{4, 5, 6, 7, 8, 9\}$.

Then $A \setminus B = \{1, 2, 3\}$. $B \setminus A = \{7, 8, 9\}$.

$A \oplus B = \{1, 2, 3, 7, 8, 9\}$.

Venn diagram



Cartesian product of sets

If A and B are two sets, the cartesian product of A and B is the set,

$$A \times B = \{(a, b) \mid a \in A, b \in B\}.$$

The cartesian ~~production~~ product of $A \times A$ is denoted as A_2.

More generally,

$$A_n = \underbrace{A \times A \times \cdots \times A}_{n \text{ times}}$$

Example:

1. Let $A = \{a, b\}$ $B = \{a, c, d\}$.

$A \times B = \{a, b\} \times \{a, c, d\}$.

$= \{(a,a)(a,c)(a,d)(b,a)(b,c)(b,d)\}$.

2. $A = \{1, 2, 3\}$ $B = \{3, 4, 5\}$.

$A \times B = \{(1,3)(1,4)(1,5)(2,3)(2,4)(2,5)(3,3)(3,4)(3,5)\}$.

$B \times A = \{(3,1)(3,2)(3,3)(4,1)(4,2)(4,3)(5,1)(5,2)(5,3)\}$

Set identities

1. Idempotent laws.

   a. $A \cup A = A$.

   b. $A \cap A = A$

2. Associative laws

   a. $(A \cup B) \cup C = A \cup (B \cup C)$

   b. $(A \cap B) \cap C = A \cap (B \cap C)$.

3. Commutative laws

   a. $A \cup B = B \cup A$.

   b. $A \cap B = B \cap A$.

4. Distributive laws

   a. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

   b. $A \cap (B \cap C) = (A \cap B) \cup (A \cap C)$.

5. Identity laws

a. $A \cup \phi = A$.

b. $A \cup U = U$

c. $A \cap U = A$.

d. $A \cap \phi = \phi$.

6. Involution laws

a. $(A^c)^c = A$.

7. Complement laws

a. $A \cup A^c = U$

b. $U^c = \phi$

c. $A \cap A^c = \phi$.

d. $\phi^c = U$

8. Demorgans laws

a. $(A \cup B)^c = A^c \cap B^c$

b. $(A \cap B)^c = A^c \cup B^c$.

Duality.

Dual of an equation is interchanging the union $\cup$ to intersection and vise versa and universal set U to empty set ϕ and vise versa.

Eg: 1. $(U \cap A) \cup (B \cap A) = A$.

$\Rightarrow (\phi \cup A) \cap (B \cup A) = A$.

2. $(A \cup B \cup C)^c = (A \cup C)^c \cap (A \cup B)^c$.

$\Rightarrow (A \cap B \cap C)^c = (A \cap C)^c \cup (A \cap B)^c$.

Problems:

1. Prove the commutative laws

a) $(A \cup B) = B \cup A$ b) $A \cap B = B \cap A$

Ans: $A \cup B = \{x : x \in A \text{ or } x \in B\} = \{x : x \in B \text{ or } x \in A\} = B \cup A$.

$A \cap B = \{x : x \in A \text{ and } x \in B\} = \{x : x \in B \text{ and } x \in A\} = B \cap A$.

2. prove the following identity : $(A \cup B) \cap (A \cup B^c) = A$.

Ans: we have i $(A \cup B) \cap (A \cup B^c) = A \cup (B \cap B^c)$ by distributive law

ii $B \cap B^c = \phi$ complement law

iii $(A \cup B) \cap (A \cup B^c) = A \cup \phi$ substitution

iv $A \cup \phi = A$. identity law

v $(A \cup B) \cap (A \cup B^c) = A$. substitution.

3. prove $(A \cup B) \setminus (A \cap B) = (A \setminus B) \cup (B \setminus A)$.

Ans: we have $X \setminus Y = X \cap Y^c$.

$\therefore (A \cup B) \setminus (A \cap B) = (A \cup B) \cap (A \cap B)^c = (A \cup B) \cap (A^c \cap B^c)$

$= (A \cap A^c) \cup (A \cap B^c) \cup (B \cap A^c) \cup (B \cap B^c)$.

$= \phi \cup (A \cap B^c) \cup (B \cap A^c) \cup \phi$

$= (A \cap B^c) \cup (B \cap A^c) = (A \setminus B) \cup (B \setminus A)$.

Generalized set operations

Consider a finite no: of sets, say $A_1, A_2, \ldots A_m$. The union and intersection of these sets are denoted and defined by

$$A_1 \cup A_2 \ldots \cup A_m = \bigcup_{i=1}^{m} A_i = \{x : x \in A_i \text{ for some } A_i\}.$$

$$A_1 \cap A_2 \ldots \cap A_m = \bigcap_{i=1}^{m} A_i = \{x : x \in A_i \text{ for every } A_i\}.$$

Theorem:

Let $\mathcal{A}$ be a collection of sets. Then

(i) $(\cup(A : A \in \mathcal{A}))^c = \cap(A^c : A \in \mathcal{A})$

(ii) $(\cap(A : A \in \mathcal{A}))^c = \cup(A^c : A \in \mathcal{A})$.

Example:

1. Consider $A_1 = \{1, 2, 3, \ldots\} = N$, $A_2 = \{2, 3, 4, \ldots\}$

$A_3 = \{3, 4, 5, \ldots\}$, $A_n = \{n, n+1, \ldots\}$.

Then

$\cup(A_n : n \in N) = N$. and $\cap(A_n : n \in N) = \phi$.

2. Let $A_i = \{i, i+1, i+2, \ldots\}$ Then

$$\bigcup_{i=1}^{n} A_i = \bigcup_{i=1}^{n} \{i, i+1, \ldots\} = \{1, 2, 3, \ldots\}.$$

$$\bigcap_{i=1}^{n} A_i = \bigcap_{i=1}^{n} \{i, i+1, i+2\} = \{n, n+1, n+2, \ldots\}.$$

The principle of Inclusion - Exclusion.

Let A and B be any finite sets. Then

$$n(A\cup B) = n(A)+n(B)-n(A\cap B).$$

Theorem:

For any finite sets A, B, C, we have

$$n(A\cup B\cup C) = n(A)+n(B)+n(C)-n(A\cap B)-n(A\cap C)-n(B\cap C)$$
$$+\,n(A\cap B\cap C).$$

Problems:

1. Find the number of mathematics students at a college taking atleast one of the languages French, German, and Russian given the following data:

65 study French	20 study French and German
45 study German	25 study French and Russian
42 study Russian	15 study German and Russian
	8 study all three languages.

Ans: we want to find $n(F\cup G\cup R)$, when F, G and R denote the set of students studying French, German and Russian resp(ective)ly.

By inclusion - exclusion principle

$$n(F\cup G\cup R) = n(F)+n(G)+n(R)-n(F\cap G)-n(F\cap R)-n(G\cap R)$$
$$+\,n(F\cap G\cap R)$$
$$= 65+45+42-20-25-15+8 = 100.$$

2. In a class of 80 students, 50 students know English, 55 know French and 46 know German language. 37 students know English and French, 28 students know French and German, 7 students know non of the languages. Find out.

a. How many students know all the three languages?

b. How many students know exactly 2 languages?

c. How many know only one language?

Ans:

$|U| = 80$

let $E \rightarrow$ set of students know English.

$|E| = 50$.

$F \rightarrow$ set of students know French.

$|F| = 55$.

$G \rightarrow$ German $|G| = 46$.

7 students know none of the languages ie

$n(E \cup F \cup G)^c = 7$.

ie $|E^c \cap F^c \cap G^c| = 7$.

$|E \cap F| = 37$ $|F \cap G| = 28$ $|E \cap G| = 25$.

$80 - |E \cup F \cup G| = 7 \Rightarrow |E \cup F \cup G| = 80 - 7 = 73$.

a) No: of students know all the three languages

ie $|E \cap F \cap G|$

we have ~~$|A \cup B \cup C| = |A| + |B| + |C|$~~

$|E \cup F \cup G| = |E| + |F| + |G| - |E \cap F| - |F \cap G| - |E \cap G| + |E \cap F \cap G|$.

ie $73 = 50 + 55 + 46 - 37 + 28 - 25 + |E \cap F \cap G|$.

$\Rightarrow |E \cap F \cap G| = \underline{\underline{12}}$ ..

b, No: of students how know exactly two languages are

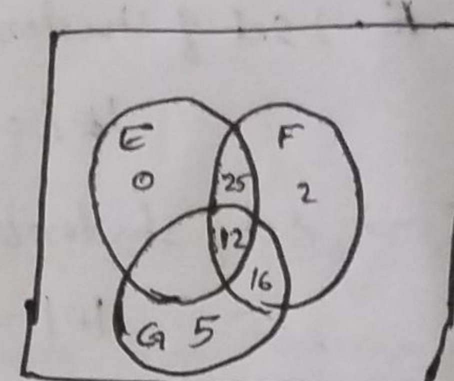
$|E \cap F| + |F \cap G| + |E \cap G| - 3 \times |E \cap F \cap G|$.

$= 37 + 28 + 25 - 36 = 54$.

c. No: of students how who know only one language.

$= |U| - |(E \cup F \cup G)^c| - |A \cap B \cap C| - |$ students know exactly 2 langu

$= 80 - 7 - 12 - 54 = \underline{7}$ ..



3. Among the integers 1 to 1000

i) How many of them are not divisible by 3 nor by 5 nor by 7.

ii) How many are not divisible by 5 or 7 but divisible by 3.

Ans: let A be set of numbers between 1 to 1000 that are divisible by 3.

B $\rightarrow$ set of numbers b/w 1 to 1000 that are divisible by 5.

C $\rightarrow$ set of " " divisible by 7.

$(a + (n-1)d)$

$|A| = |1000/3| = 333$.

$|B| = |1000/5| = 200$.

$|C| = |1000/7| = 142$.

$|A \cap B \cap C| = \left| 1000/3\times5\times7 \right| = 9.$

$|A \cap B| = \left| 1000/3\times5 \right| = 66.$

$|B \cap C| = \left| 1000/7\times5 \right| = 28.$

$|A \cap C| = \left| 1000/3\times7 \right| = 47.$

i. Cardinality of set of numbers that are not divisible by 3 nor by 5 nor by 7 i.e

$\cancel{|A^c \cap B^c \cap C^c|} = |(A \cup B \cup C)^c| = |U| - |A \cup B \cup C|$

$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |A \cap C| + |A \cap B \cap C|$

$= 333 + 200 + 142 - 66 - 47 - 28 + 9.$

$= 543.$

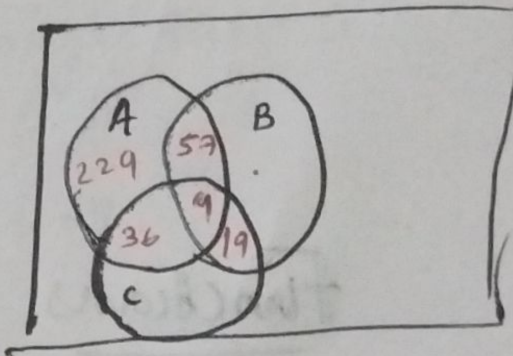
$\therefore |A^c \cap B^c \cap C^c| = 1000 - 543 = \underline{\underline{457}}.$

ii. Cardinality of set of numbers that are not divisible by 5 and 7 but divisible by 3 is

$A \cap B^c \cap C^c.$

$= |A \cap (B \cup C)^c|$

$= |A - (\overline{B \cup C})|$ Consider venn diagram



$= 333 - (57 + 9 + 38) = \underline{\underline{229}}.$

4. Among 50 students in a class, 26 got an A in the first examination and 21 got an A in the second examination. If 17 students did not get an A in either examination. how many students got A in both the examination.

Ans: $|U| = 50$.

$E_1 \to$ Set of students for A in first examination

$E_2 \to$ " 2nd exam

$|E_1| = 26$

$|E_2| = 21$.

$\therefore |E_1^c \cap E_2^c| = |(E_1 \cup E_2)^c| = 17$.

$|E_1 \cup E_2| = 50 - 17 = 33$

$\therefore$ ~~$E_1 A$~~ we have

$$|E_1 \cup E_2| = |E_1| + |E_2| - |E_1 \cap E_2|.$$

$$\therefore |E_1 \cap E_2| = |E_1| + |E_2| - |E_1 \cup E_2|$$

$$= 26 + 21 - 33 = 14.$$

Functions.

Definition:

Suppose that to each element of a set A we assign a unique element of a set B, the collection of such assignment is called a function from A to B. The set A is called the domain of the function and the set B is called the co-domain of the function.

we write $f: A \to B$.

* functions are sometimes also called mappings or transformation.

* If $f(a) = b$, we say that b is the image of a and a is the ~~fact~~ preimage of b.

* The range of f is the set of all images of elements of A.

Eg:

1. Let $f: Z \to Z$ assign the square of an integer ~~that to~~ this integer. Then $f(x) = x^2$, where the domain of f is the set of all integers.

Co-domain - Z.

Range - $\{0, 1, 4, 9, \ldots\}$.

2. Floor function, ceiling function

Let x be any real number,

$\lfloor x \rfloor$ is called the floor of x, denote the greatest integer that does not exceed x.

$\lceil x \rceil$, called the ceiling of x, denotes the least integer that is not less than x.

Eg: $\lfloor 3.14 \rfloor = 3$. $\lfloor \sqrt{5} \rfloor = 2$. $\lfloor -8.5 \rfloor = -9$.

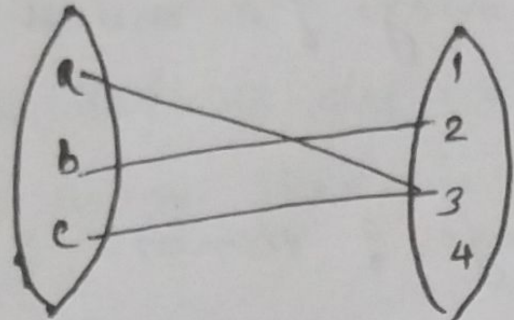
$\lceil 3.14 \rceil = 4$. $\lceil \sqrt{5} \rceil = 3$. $\lceil -8.5 \rceil = -8$

Domain : R.

Co-domain : R.

Range : Z.

$f : A \rightarrow B$



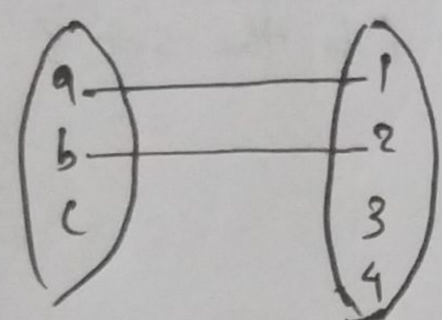
Domain : $\{a, b, c\}$.

Range : $\{2, 3\}$.

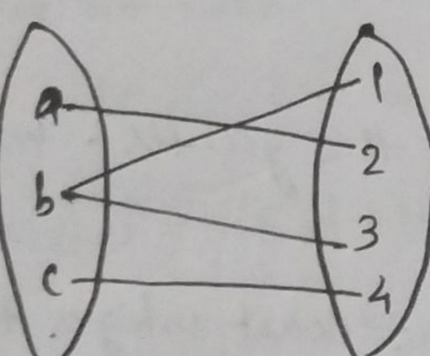
Codomain : $\{1, 2, 3, 4\}$.

A function.

$f : A \rightarrow B$



Not a function.



Not a function.

one-to-one function. (Injective)

If $f : A \rightarrow B$ is one-to-one if and only if $\forall\, a_1, a_2 \in A$

$f(a_1) = f(a_2) \Rightarrow a_1 = a_2$

Eg:

1. $A = \{1, 2, 3\}$ $B = \{1, 2, 3, 4, 5\}$.

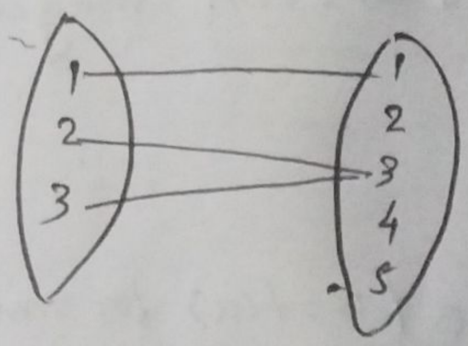
$F = \{(1,1)\ (2,3)(3,4)\}$.

$f(1) = f(1) \Rightarrow 1 = 1$

$f(2) = f(2) \Rightarrow 2 = 2.$

$f(3) = f(3) \Rightarrow 3 = 3.$

2. $f = \{(1,1)\ (2,3)\ (3,3)\}.$



$f(1) = f(1) \Rightarrow 1 = 1$

$f(2) = f(3) \Rightarrow 2 \neq 3.$

function is not one-one.

3. $f: R \to R$ be a function defined by $f(x) = 3x + 7$,

Check whether f is one-to-one.

Ans: $f(x_1) = f(x_2) \Rightarrow x_1 = x_2.$

LHS:

Let $f(x_1) = f(x_2)$

$\Rightarrow 3x_1 + 7 = 3x_2 + 7.$

$\Rightarrow 3x_1 = 3x_2$

$\Rightarrow x_1 = x_2.$

$\therefore$ function is one-to-one.

4. $f: Z \to Z$ be a function defined by $f(x) = 2x$

check whether f is one-to-one and find ~~image~~ Range.

~~Ans:~~

Ans: $f(x_1) = f(x_2)$

$\Rightarrow 2x_1 = 2x_2$

$\Rightarrow x_1 = x_2$

$\therefore$ f is one-to-one.

Range $= \{2x \mid x \in Z\} = \{2, 4, \ldots\} =$ even integers.

* <u>Image</u>.

If $f: A \to B$ and $A_1 \subseteq A$ then $f(A_1) = \{b \in B \mid b = f(a)$ for some $a \in A\}$

is called the image of A under f.

1. Let $A = \{1, 2, 3, 4, 5\}$ $B = \{x, y, z, w\}$ Let $f: A \to B$ is given by $f = \{(1,w)(2,x)(3,x),(4,y)(5,y)\}$. and

$A_1 = \{1\}$. $A_2 = \{1, 2\}$ $A_3 = \{1, 2, 3\}$ $A_4 = \{2, 3\}$.

$\Rightarrow f(A_1) = f(1) = \{w\}$.

$f(A_2) = \{f(1), f(2)\} = \{w, x\}$.

$f(A_3) = \{w, x\}$.

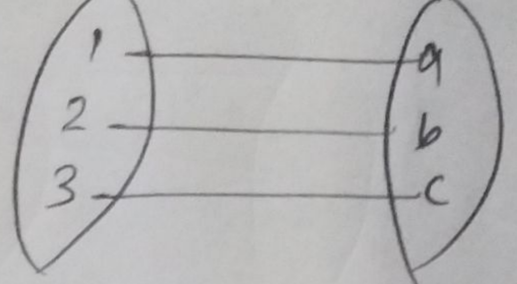
$f(A_4) = \{x\}$.

onto - functions (Surjective)

A function $f : A \to B$ is said to be 'onto' if $\forall\, b \in B$

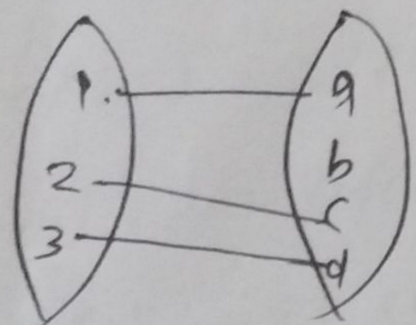
There is atleast one $a \in A$ $\therefore$ $f(a) = b$.

Eg: $f : A \longrightarrow B$



here all the elements in B has preimage.

2. $f : A \to B$



It is not onto.

Restrictions:

If $f : A \to B$ and $A_1 \subseteq A$, then $f|_{A_1} : A_1 \to B$ is called restriction of f! if $f|_{A_1}(a) = f(a)\ \forall\, a \in A_1$.

Extensions:

$f : A_1 \to B$ and $A_1 \subseteq A$ and $g : A \to B$, $g(a) = f(a)\ \forall\, a \in A_1$,

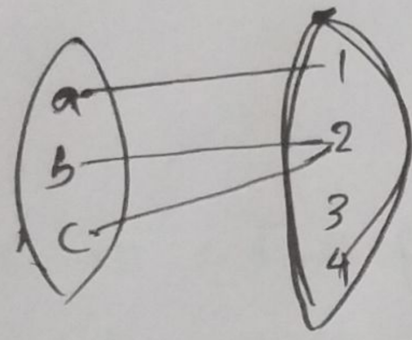
then ~~g is~~ called g an extension of f.

Byective function:

A function $f: A \to B$ which is both one-to-one and onto is called bijection.

Eg:

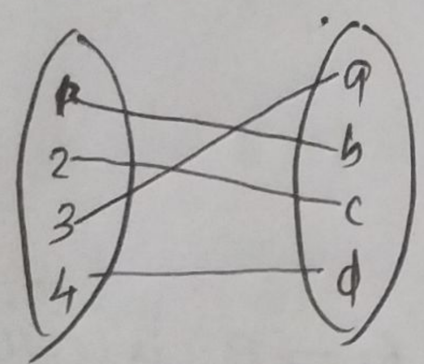
$f: A \to B$



Not byection

$\because$ f is not one-~~to~~ one. , not onto.

2. $f: A \to B$



f is a byection.

Composition of functions:

If $f: A \to B$ and $g: B \to C$, define the composite function $g \circ f: \cancel{A} \; A \to \cancel{C}$ by $(g \circ f)(a) = g(f(a))$.

Q1: Let $A = \{1, 2, 3, 4\}$ $B = \{a, b, c\}$, $C = \{w, x, z\}$.
with $f: A \to B$ and $g: B \to C$ given by
$f = \{(1, a), (2, a)(3, b)(4, c)\}$ and $g = \{(a, x), (b, y)(c, z)\}$,
Find the relation $g \circ f$.

Ans: $g \circ f : A \to C$.

$\cancel{gf(1) =}$
$g \circ f(1) = g(f(1)) = g(a) = x$.
$g \circ f(2) = g(f(2)) = g(a) = x$.
$g \circ f(3) = g(f(3)) = g(b) = y$.
$g \circ f(4) = g(f(4)) = g(c) = z$.
$\therefore g \circ f = \{(1, x)(2, x)(3, y)(4, z)\}$.

Q2: Let $f: R \to R$ and $g: R \to R$ by $f(x) = x^2$, $g(x) = x + 5$.
Find $(f \circ g)(x)$ and $(g \circ f)(x)$.

Ans: $f \circ g : R \to R$
$f \circ g(x) = f(g(x)) = f(x+5) = (x+5)^2$.
$g \circ f(x) = g(f(x)) = g(x^2) = x^2 + 5$.

Q_3: Let the functions f and g be defined by $f(x) = 2x+1$ and $g(x) = x^2 - 2$. Find the formula defining the composition function $g \circ f$.

Ans: $g \circ f(x) = g(f(x)) = g(2x+1) = (2x+1)^2 - 2$
$$= 4x^2 + 4x - 1.$$

Inverse function.

Let $f : A \to B$ be a function, then the inverse of f is $f^{-1} : B \to A$.

Theorem:
A function $f : A \to B$ is invertible if and only if f is both one-to-one and onto.

Q_1: Let $f : R \to R$ be defined by $f(x) = 2x - 3$. Now f is one-one and onto; hence f has an inverse function f^{-1}. Find a formula for f^{-1}.

Ans: Check for ~~onto~~ one-to-one and onto.

Let $y = f(x) = 2x - 3$.

ie $y = 2x - 3$.

$\Rightarrow 2x = y + 3$.

$\Rightarrow x = \frac{y+3}{2}$

$\therefore f^{-1}(y) = \frac{y+3}{2}$ replace y by x.

$\therefore f^{-1}(x) = \frac{x+3}{2}$.

2. $f: A \longrightarrow B$.

Let $A = \{a, b, c\}$ $B = \{1, 2, 3\}$.

$f(a) = 2$, $f(b) = 3$, $f(c) = 1$.

$\therefore$

$f^{-1}(1) = c$, $f^{-1}(2) = a$, $f^{-1}(3) = b$.

Q3: Let $f: Z \to Z$ $\therefore f(x) = x+1$. Is f is invertible?
If yes, find inverse.

Ans: one-to-one.

$f(x_1) = f(x_2)$

$\Rightarrow x_1 + 1 = x_2 + 1$

$\Rightarrow x_1 = x_2$.

$\therefore$ one-to-one.

also onto $\therefore$ for every element $a \in Z$ $\exists$ an element $b \in Z$
$\therefore f(b) = a$.

Now to find f^{-1}

let $y = f(x) = x+1$

$y = x+1$

$\Rightarrow x = y-1$

$\therefore f^{-1}(y) = y-1$

$\therefore f^{-1}(x) = x-1$.

Cardinality of sets.

Two sets A and B are said to be equipotent, or to have the same number of elements or the same cardinality written $A \simeq B$, If there exist a one-to-one correspondence $f: A \to B$. A set A is finite If A is empty or If A has the same cardinality as the set $\{1, 2, \dots n\}$ for some positive integer n. A set is infinite If it is not finite.

Eg Example for infinite sets: $N, Z, R, Q \dots$

We now introduce the idea of 'Cardinal numbers', we will consider cardinal numbers simply as symbols of assigned to set in such a way that two sets are assigned the same symbol If and only If they have the same cardinality. denoted by $|A|$, $n(A)$ or card (A).

$|N| = \aleph_0$ (aleph-naught). (Introduced by Cantor)

Example:

1. $|(x, y, z)| = 3$

2. $|\{1, 3, 5, 7, 9\}| = 5$.

3. Let $E = \{2, 4, 6, 8, \dots\}$, the set of even +ve integers. The function $f: N \to E$ defined by $f(n) = 2n$ is a one-to one correspondence between the positive integers N and E. thus $|E| = |N| = \aleph_0$.

* A set with cardinality $\aleph_0$ is said to be denumerable or countably infinite.

Theorem:

A countable union of countable set is countable.

Q1: Find the cardinal number of each of the sets:

a. $A = \{a, b, c, \ldots z\}$.

b. $B = \{1, -3, 5, 11, -28\}$

c. $C = \{x : x \in N, x^2 = 5\}$

d. $D = \{10, 20, 30, \ldots\}$

e. $E = \{6, 7, 8, 9, \ldots\}$

Ans:

a) $|A| = 26$ no: of letters in English alphabet.

b) $|B| = 5$.

c) $|C| = 0$ $\because$ No +ve integers $\ni$ $x^2 = 5$.

d) $|D| = \aleph_0$ $\because$ $f(n) = 10n$ has 1-1 correspondence with N.

e) $|E| = \aleph_0$ $\because$ $g : N \to E$ by $g(n) = n+5$. 1-1.

Cantor diagonalization Argument.

Theorem:

~~The set $\{0,1\}^\infty$ is an uncountable set.~~

- $\{0,1\}^\infty$ · set of binary strings of infinite length.

$x = 00000000\ldots$

$y = 010101010101\ldots$

$z = 0110101001010 10\ldots \quad z_n = 1$ iff n is prime.

It has infinite no: of strings.

Theorem:

The set $\{0,1\}^\infty$ is an uncountable set

Proof:

Proof is by contradiction.

Assume that the set $\{0,1\}^\infty$ is countable with the sequencing

$r_1, r_2, \ldots r_n \ldots$

$r_1 = d_{11} d_{12} d_{13} \ldots\ldots$

$r_2 = d_{21} d_{22} d_{33} \ldots\ldots$

$r_3 = d_{31} d_{32} d_{33} d_{34} \ldots\ldots$

$\vdots$

we are going to show that $\exists$ a ~~sequence~~ string which is of infinite length and binary, but not contained in $\{0,1\}^\infty$.

Consider the diagonal binary string $r = d_{11} d_{22} d_{33} \ldots d_{nn} \ldots$

$\bar{r} = \bar{d}_{11} \bar{d}_{22} \bar{d}_{33} \ldots \bar{d}_{nn} \ldots$

the string $\bar{r}$ ~~$\notin \{0,1\}^\infty$ and~~ is different from all the strings $r_1, r_2, \ldots r_n, \ldots$

and ~~in the sequ~~ $\bar{r}$ is not contained in the sequencing of strings in $\{0,1\}^{\alpha}$.

$\therefore \exists$ a contradiction.

$\Rightarrow \{0,1\}^{\infty}$ is uncountable set.

[$\{0,1\}^*$ is an infinite set with binary strings ~~but~~ in which the length of the string is finite.]

* Cantor's diagonalization is not applicable to prove ~~that~~ $\{0,1\}^*$ is an uncountable set.

* " $\mathbb{Z}$ is an uncoun- (each integer has finite digit decimal)

" $\mathbb{Q}$.

Theorem:

The set of all real numbers between 0 and 1 is uncountable.

Cantor's theorem

For every set A $|A| < |P(A)|$.

Proof: true for finite sets

ie $|A| = n \Rightarrow |P(A)| = 2^n$.

Now for an infinite set proof is by contradiction.

~~on tot~~ If $|A| \geq |P(A)|$ ($\because |X| \geq |Y|$, then $\exists$ a surjection from X to Y.)

ie $\exists$ a surjection ~~from~~ $f: A \to P(A)$.

let .

	x_1	x_2	x_3	...	
$f(x_1)$	$\underline{0}$	0	1	1 ...	1
$f(x_2)$	1	$\underline{1}$	1	...	0
$f(x_3)$	1	1	$\underline{0}$	...	1
$\vdots$			$\vdots$		

we construct a set S $\therefore$ S is different form all $f(x_i)$.

Consider the diagonal entries form the table,

$S = \{x_1, x_3, \ldots\}$

$\therefore S \in P(A)$.

$S \neq f(x_i)$ for all $x_i \in A$.

$\Rightarrow \nexists$ a surjection from $A \to P(A)$

$\Rightarrow |A| < |P(A)|$.

Relations

Let A and B be sets, A binary relation from A to B is a subset of $A \times B$.

We use the notation aRb to denote $(a,b) \in R$ and $a \not R b$ to denote that $(a,b) \notin R$.

Eg: Let $A = \{0,1,2\}$ and $B = \{a,b\}$. Then $\{(0,a)\ (0,b)\ (1,a)\ (2,b)\}$ is a relation form A to B.

2. Let A be the set $\{1, 2, 3, 4\}$. Which ordered pair are in the relation $R = \{(a,b) \mid a \text{ divides } b\}$.

Ans:

$$R = \{(1,1)(1,2)(1,3)(1,4)(2,2)(2,4)(3,3)(4,4)\}.$$

3. Consider these relations on the set of integers.

$R_1 = \{(a,b) \mid a \le b\}$

$R_2 = \{(a,b) \mid a > b\}$

$R_3 = \{(a,b) \mid a = b \text{ or } a = -b\}$

$R_4 = \{(a,b) \mid a = b\}$.

$R_5 = \{(a,b) \mid a = b+1\}$

$R_6 = \{(a,b) \mid a+b \le 3\}$.

Which of these relations contains each of the pairs $(1,1)(1,2)(2,1)(1,-1)$ and $(2,2)$.

Ans: $(1,1)$ is in R_1, R_3, R_4, R_6.

$(1,2)$ is in R_1, R_6.

$(2,1)$ is in R_2, R_5, R_6.

$(1,-1)$ is in R_2, R_3, R_6.

$(2,2)$ is in R_1, R_3, R_4.

Result.

For any finite set A and B with $|A| = m$, and $|B| = n$, then there are 2^{mn} relations from A to B including empty set and $A \times B$ itself.

Result

1. $A \times \phi = \phi$.
2. $A \times (B \cap C) = (A \times B) \cap (A \times C)$
3. $A \times (B \cup C) = (A \times B) \cup (A \times C)$
4. $(A \cap B) \times C = (A \times C) \cap (B \times C)$.
5. $(A \cup B) \times C = (A \times C) \cup (B \times C)$.

Q. If $A = \{1, 2, 3, 4\}$ $B = \{2, 5\}$. $C = \{3, 4, 7\}$

a. Find $A \cup (B \times C) = (A \cup B) \times (A \cup C)$.

b. $(A \cup B) \times C = (A \times C) \cup (B \times C)$

Properties of relation

1. Reflexive : A relation R on a set A is called reflexive if $(a, a) \in R$ $\forall a \in A$.

2. Symmetric : If ~~(a,b)∈R~~ $(b, a) \in R$ whenever $(a, b) \in R$ $\forall a, b \in A$.

3. Transitive : If $(a, b) \in R$ and $(b, c) \in R$, then $(a, c) \in R$

Eg:

1. Consider the following relations on $\{1, 2, 3, 4\}$.

$R_1 = \{(1,1) (1,2) (2,1) (2,2) (3,4) (4,1) (4,4)\}$.

$R_2 = \{(1,1) (1,2) (2,1)\}$

$R_3 = \{(1,1) (1,2) (1,4) (2,1) (2,2) (3,3) (4,1) (4,4)\}$.

Which of these relations satisfies reflexive, symmetry, transitive.

R_3.

2. Is the 'divides' relation on the set of positive integers is reflexive, symmetric and transitive?

Ans: In divides, let $a \in \mathbb{Z}$, then $a|a \ \forall a \in \mathbb{Z}$.

$\therefore$ ~~dis~~ reflexive.

Now if $a|b \not\Rightarrow b|a$ Eg: $2|4$, but $4 \nmid 2$.

$\therefore$ Not Symmetric

If $a|b$ and $b|c \Rightarrow a|c$.

Eg: $2|4$, $4|8 \Rightarrow 2|8$.

$\therefore$ transitive.

Properties

4. Anti symmetric $\Rightarrow$: $\forall\ a,b \in A$, $(a,b) \in R$ and $(b,a) \in R$
$\Rightarrow a = b$.

5. Irreflexive relation : $\forall\ a \in A$, $(a,a) \notin A$.

6. Equivalence relation : A relation R on a set A is said to be equivalence relation if R is ~~symm~~ reflexive, symmetric and transitive.

7. partial ordering relation : A relation R on a set A is said to be partial ordering if R is reflexive ~~and~~ antisymmetric and transitive.

Composition

Let R be a relation from ~~A to~~ a set A to a set B and S a relation from B to C. Then Composite of R and S is the relation consisting of ordered pairs (a,c), when $a \in A$ and $c \in C$, and for which $\exists$ an element $b \in B$ $\therefore$ $(a,b) \in R$ and $(b,c) \in S$. We denote the composite of R and S ~~as~~ by $S \circ R$.

Eg:
1. What is the composite of the relations R and S, where R is the relation from $\{1,2,3\}$ to $\{1,2,3,4\}$, with $R = \{(1,1)(1,4)(2,3)(3,1)(3,4)\}$ and S is the relation from $\{1,2,3,4\}$ to $\{0,1,2\}$ with $S = \{(1,0)(2,0)(3,1)(3,2)(4,1)\}$.

Ans: $S \circ R = \{(1,0)(1,1)(2,1)(2,2)(3,0)(3,1)\}$.

* ~~The~~ Let R be a relation on the set A. The powers $R^n, n = 1, 2, \ldots$ are defined recursively by $R^1 = R$, $R^{n+1} = R^n \circ R$.

Eg: $R^2 = R \circ R$, $R^3 = R^2 \circ R, \ldots$

n-ary Relations.

Definition: Let $A_1, A_2, \ldots A_n$ be sets. An n-ary relation on these sets is a subset of $A_1 \times A_2 \times \cdots \times A_n$. The sets $A_1, A_2, \ldots A_n$ are called the domains of the relation. and 'n' is called its degree.

Eg: 1. Let R be the relation on $N \times N \times N$ consisting of triples (a,b,c) where a, b, and c are integers with $a < b < c$. Then $(1,2,3) \in R$ but $(2,4,3) \notin R$.

degree = 3.

Its domains are set of all natural numbers.

2. Let L be a line in the plane. Then 'betweenness' is a ~~set~~ ternary relation R on the points of L. i.e $(a,b,c) \in R$ if b lies between a and c on L.

Operations on n-ary relation

1. Let R be an n-ary relation and C a condition that elements in R may satisfy. Then the selection operator S_c maps the n-ary relation R to the n-ary relation of all n-tuples from R that satisfy the condition C.

2. The projection $P_{i_1, i_2, \dots i_m}$ where $i_1 < i_2 \dots < i_m$ maps the n-tuple $(a_1, a_2 \dots a_n)$ to the m-tuple $(a_{i_1}, \dots a_{i_m})$ where $m \leq n$.

3. Let R be a relation of degree m and S a relation of degree n. The ~~joint~~ Join $J_p(R, S)$. where $p \leq m$ and $p \leq n$ is a relation of degree $m+n-p$ that consist of all $(m+n-p)$-tuple $(a_1, \dots a_{m-p}, c_1, c_2, \dots c_p, b_1, b_2, \dots b_{n-p})$. where the m-tuple $(a_1, a_2 \dots a_{m-p}, c_1, c_2, \dots c_p)$ belongs to R and the n-tuple $(c_1, c_2 \dots c_p, b_1, b_2, \dots b_{n-p})$ belongs to S.

Representing relations using Matrices

A relation between finite sets can be represented using a zero-one matrix. Suppose that R is a relation from $A = \{a_1, a_2 \ldots a_m\}$ to $B = \{b_1, \ldots \ldots b_n\}$. The relation R can be represented by the matrix $M_R = [m_{ij}]$, where

$$m_{ij} = \begin{cases} 1 & \text{if } (a_i, b_j) \in R. \\ 0 & \text{if } (a_i, b_j) \notin R \end{cases}$$

i.e the zero-one matrix representing R has a 1 as its (i, j) th entry when a_i is related to b_j, and a 0 in the position if a_i is not related to b_j.

Example :

1. Suppose that $A = \{1,2,3\}$ and $B = \{1,2\}$. Let R be the relation from A to B containing (a,b) if $a \in A$ and $b \in B$ and $a > b$. What is the matrix representing R if $a_1 = 1$ $a_2 = 2$ and $a_3 = 3$ and $b_1 = 1$, $b_2 = 2$.

Ans: $R = \{(2,1)\ (3,1)\ (3,2)\}$.

$$\therefore M_R = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}$$

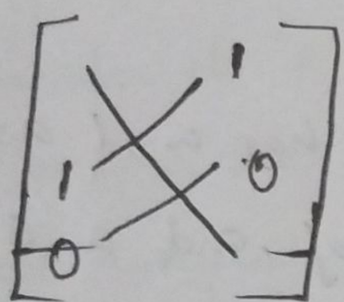
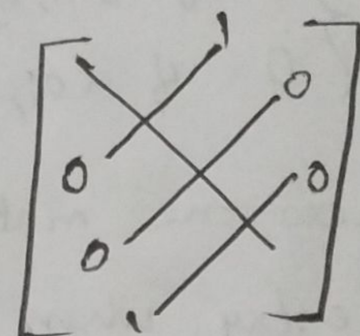
2. Let $A = \{a_1, a_2, a_3\}$ and $B = \{b_1, b_2, b_3, b_4, b_5\}$. which ordered pair are in the relation R representing the matrix

$$M_R = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \end{bmatrix}$$

Ans: $R = \{(a_1, b_2)(a_2, b_1)(a_2, b_3)(a_2, b_4)(a_3, b_1)(a_3, b_3)$
$(a_3, b_5)\}$.

2.

* In the given matrix if all the diagonal elements are 1, then R is reflexive.

*  Symmetric  Antisymmetric

3. Suppose that the relation R on a set is represented by the matrix

$$M_R = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

is R reflexive symmetric and or antisymmetric ?

Ans: All the diagonal entries are 1. $\therefore$ R is reflexive

Also R is symmetric

or $R = \{(1,1)(1,2)(2,1)(2,2)(2,3)(3,2)(3,3)\}$.

$\therefore$ for R, reflexive, symmetric.

4. Suppose that the relations R_1 and R_2 on a set A are represented by the matrices

$$M_{R_1} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \qquad M_{R_2} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

What are the matrices representing $R_1 \cup R_2$ and $R_1 \cap R_2$.

Ans: $M_{R_1 \cup R_2} = M_{R_1} \vee M_{R_2} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

$M_{R_1 \cap R_2} = M_{R_1} \wedge M_{R_2} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

5. Find the matrix representing the relations $S \circ R$, where the matrices representing R and S are

$$M_R = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \text{ and } M_S = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

Ans: $M_{S \circ R} = M_R \odot M_S = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ $\quad t_{ij} = 1$ if $r_{ik} = s_{kj} = 1$

Equivalence relation

Consider a non-empty set A is ~~called~~, a relation R on A is an equivalence relation, if R is reflexive, symmetric, and transitive.

* two elements a and b that are related by an equivalence relation are called equivalent. denoted by $a \sim b$.

Example:

1. Let R be the relation on the set of Integers s.t. aRb iff $a = b$ or $a = -b$. Show that R is reflexive, symmetric and transitive.

Ans:

$a = b \Rightarrow a = a$. $(a, a) \in R$.

$\therefore$ reflexive.

$a = b \Rightarrow b = a$ symmetric

$a = b$, $b = c \Rightarrow a = c$ transitive.

$\therefore$ Equivalence relation.

2. Congruence modulo m Let m be a +ve integer with $m > 1$ Show that the relation $R = \{(a, b) \mid a \equiv b \pmod{m}\}$. is an equivalence relation on the set of Integers.

Ans: $a \equiv b \bmod m$ iff m divides $a - b$.

$a - a = 0$ is divisible by m. $\because 0 = 0 \cdot m$.

$\therefore$ reflexive.

If aRb ie a-b is divisible by m $\Rightarrow$ $b-a = -(a-b)$ is divisible by m.

$\therefore$ ie $b \equiv a \mod m$

$\therefore$ Symmetric

If aRb ie $a \equiv b \mod m$ and $b \equiv c \mod m$.

ie $a-b = km$, $b-c = cm$.

$\therefore$ $a-c = (k+c)m$.

$\therefore a \equiv c \mod m$.

$\therefore$ transitive.

$\therefore$ R is an equivalence relation.

Equivalence classes and Partition.

Defi: Let R be an equivalence relation on a set A. The set of all elements that are related to an element a of A is called an equivalence class of a. The equivalence class of a w.r.t R is denoted by $[a]_R$ or $[a]$

ie $[a] = \{b \mid (a,b) \in R\}$.

If $b \in [a]$, then b is called a representative of the equivalence class.

Eg:
1. Define the relation R on the set Z by aRb if $a^2 = b^2$. find the equivalence classes.

Ans: $[0] = \{0\}$

$[1] = \{1, -1\}$

$[2] = \{-2, 2\}$.

Def: Given a set A and index set I. Let $\phi \neq A_i \subseteq A$ for each $i \in I$. Then $\{A_i\}_{i \in I}$ is a partition of A if

a) $A = \bigcup_{i \in I} A_i$ and

b) $A_i \cap A_j = \phi \quad \forall\, i, j \in I \quad i \neq j$.

Example :

1. If $A = \{1, 2, 3, \ldots 10\}$ then determine a partition of A.

a) $A_1 = \{1, 2, 3, 4, 5\}$, $A_2 = \{6, 7, 8, 9, 10\}$

b) $A_1 = \{1, 2, 3\}$ $A_2 = \{4, 6, 7\}$ $A_3 = \{5, 8, 9, 10\}$.

Partial ordering.

Theorem :

Let R be an equivalence relation on a set A. These statements for elements a and b of A are equivalent

(i) aRb (ii) $[a] = [b]$ (iii) $[a] \cap [b] \neq \phi$.

Partial ordering.

A relation R on a set A is called a partial order relation if R is reflexive, antisymmetric and transitive.

Example :

1. Check whether the relation defined on Z by aRb if $a \leq b$ is partial order relation ?

Ans: We have

$a \leq a$. $\Rightarrow$ aRa ie R is reflexive

$a \leq b \not\Rightarrow b \leq a$.

If $a \leq b$ and $b \leq a \Rightarrow a = b$.

ie R is antisymmetric

$a \leq b$ and $b \leq c \Rightarrow a \leq c$.

$\therefore$ R is transitive.

$\therefore$ R is a partial order.

H.W.

1. Show that the Inclusion relation $\subseteq$ is a partial ordering on the power set of a set S.

definition: The elements a and b of a poset $(S, \preceq)$ are called Comparable if either $a \preceq b$ or $b \preceq a$. otherwise incomparable.

- If $(S, \preceq)$ is a poset and every two elements of S are comparable, S is called ~~tot~~ totally ordered or linearly ordered set, and $\preceq$ is called a total order or a linear order. A totally ordered set is also called a chain.

Eg:
1. $(Z, \leq)$ is totally ordered.
2. $(Z^+, |)$ is not totally ordered.

- $(S, \preceq)$ is a well-ordered set if it is a poset $\therefore \preceq$ is a ~~totally ordering~~ total ordering and every non-empty Subset of S has a least element.

Hasse Diagrams

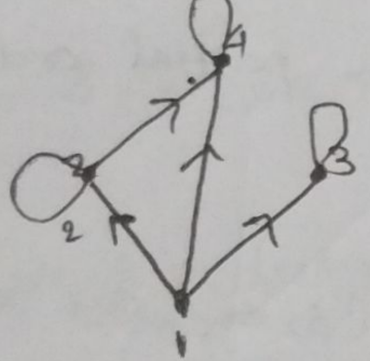
If R is a partial order on a finite set A, we construct a Hasse diagram for R on A by drawing a line segment from x upto y, if $x, y \in A$ $\therefore$ xRy.

Example:

1. $A = \{1, 2, 3, 4\}$ R is defined on A $\therefore$ xRy if $x | y$.

Then $R = \{(1,1)(1,2)(1,3)(1,4)(2,2)(2,4)(3,3)(4,4)\}$.

we can draw a graph like

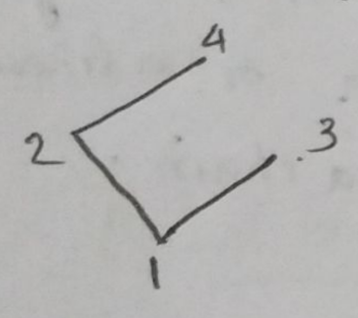


But since all posets are reflexive, we avoid the loops.

Since by transitive relation $1R2$ and $2R4 \rightarrow 1R4$.

we avoid the line from 1 to 4.

Hasse diagram



2. Draw the Hasse diagram for the following 4 posets

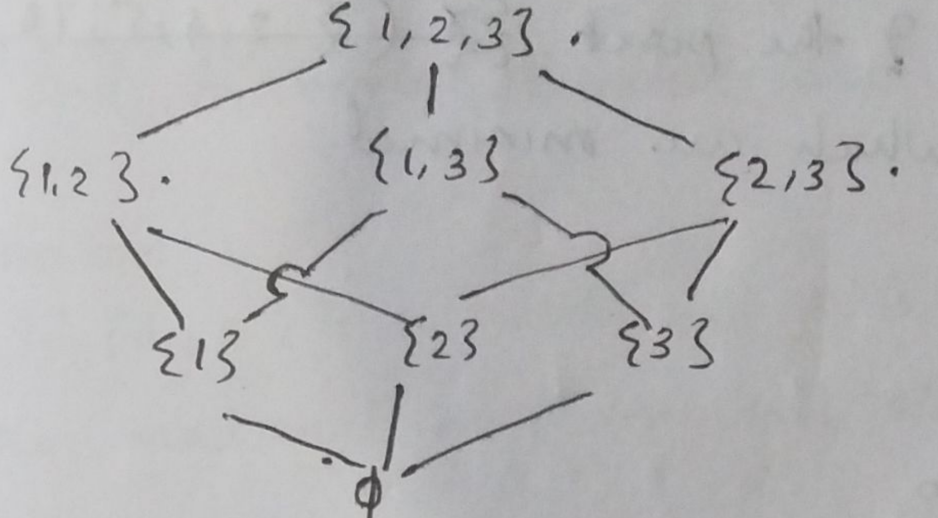
a) $U = \{1, 2, 3\}$ $A = P(U)$ R is the subset relation on A.

b, R is the exactly divides relation applied to $A = \{1, 2, 4, 8\}$.

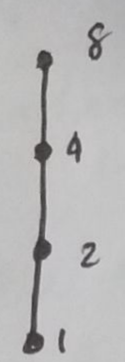
c. R is the exactly divides relation applied to $A = \{2, 3, 5, 7\}$.

Ans:

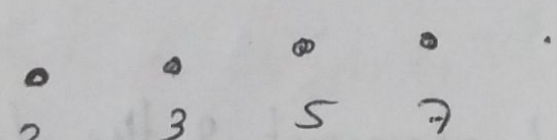
a/ $U = \{1, 2, 3\}$ $A = p(u) = \{\phi, \{1\}, \{2\}, \{3\}, \{1,2\}, \{2,3\}, \{1,3\}, \{1,2,3\}\}$.



b, $R = \{(1,2)(1,4)(1,8)(2,4)(2,8)(4,8)(1,1)(2,2)(4,4)(8,8)\}$



c. $R = \{(2,2)(3,3)(5,5)(7,7)\}$



Maximal and minimal elements:

An element of a poset is called maximal if it is not less than any elements of the poset. i.e a is maximal in the poset $(S, \preceq)$ if there is no $b \in S$ s.t. $a \prec b$.

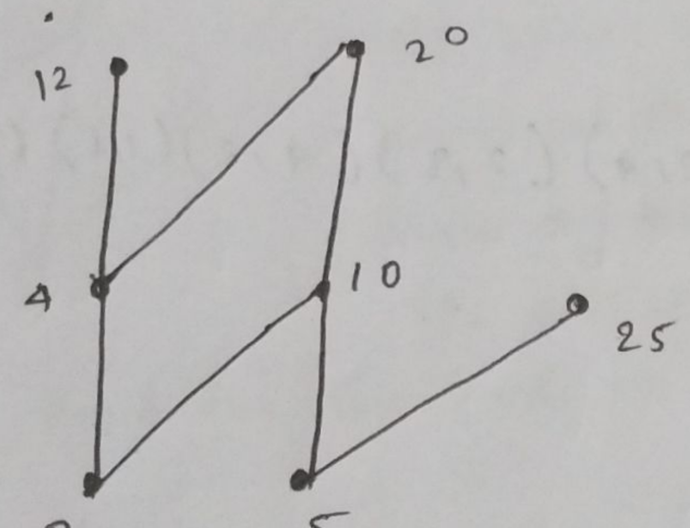
An element of a poset S is called a minimal S

If it is not greater than any element of the poset. i.e. a minimal of if there is no element $b \in S$ s.t. $b < a$.

Example:

1. Which elements of the poset ~~{~~ $(\{2, 4, 5, 10, 12, 20, 25\}, |)$ are maximal, and which are minimal.

~~R = {~~



from the diagram 12, 20 and 25 are maximal.

2 and 5 are minimal.

2. Find the maximal and minimal element of the following

a) Let $A = \{1, 2, 4, 8\}$ and R is exactly divides

b) Let $B = \{2, 3, 5, 7\}$,,

Ans: We have divides is a poset.

a, $R = \{(1,1)\ (1,2)(1,4)(1,8)\ (2,4)(2,8)\ (4,8)\ (2,2)(4,4)\ (8,8)\}$.

Minimal element is 1

maximal element is 8.

b) $R = \{(2,2)(3,3)(5,5)(7,7)\}$.

Minimal elements are 2, 3, 5, 7

and maximal elements are 2, 3, 5, 7.

Theorem :

If (A, R) is a poset and A is finite then A has both the maximal and minimal elements.

Defn: If (A, R) is a poset, then an element $x \in A$ is called a least element if $x R a$ for all $a \in A$.

element $y \in A$ is called a greatest element if $a R y$ $\forall a \in A$.

Eg: Let $U = \{1, 2, 3\}$ and let R be the subset relation.

Find the least and greatest element.

Ans: $P(U) = \{\phi, \{1\}, \{2\}, \{3\}, \{1,2\} \{1,3\}, \{2,3\}, \{1,2,3\}.\}$.

least element $= \phi$

greatest element $= U = \{1, 2, 3\}$.

Theorem :

If the poset (A, R) has a greatest (least element), then the element is unique.

2. For poset given below find the least and greatest elements.

a) $A = \{1, 2, 4, 8\}$ xRy $x|y$.

Ans: least element is 1 $\because$ $1/1, 1/2, 1/4, 1/8$.

greatest element is 8 $\because$ $1/8, 2/8, 4/8, 8/8$.

Lattices

Def: A lattice is a partially ordered set (poset) (A, R) in which for every pair of element $a, b \in A$, the least upper bound, lub $\{a, b\}$ and the greatest lower bound glb $\{a, b\}$ both exists in A.

We denote lub $\{a, b\}$ by $a \vee b$ or $a \oplus b$ or $a + b$ (join)

" glb $\{a, b\}$ by $a \wedge b$ or $a * b$ or $a \cdot b$ (meet)

So we define a lattice (A, R) by $\langle A, +, \cdot \rangle$ or

$\langle A, \oplus, * \rangle$ or

$\langle A, \vee, \wedge \rangle$.

Def: Let (A, R) be a poset with $B \subseteq A$. An element $x \in A$ is called a lowerbound of B if xRb for all $b \in B$.

An element $y \in A$ is called an upperbound of B if bRy $\forall b \in B$.

An element $x' \in A$ is called greatest lower bound glb of B if it is a lower bound of B and if to all other lower bounds $x'' \in R$ we have $x'' R x'$.

iii) $y' \in A$ is a least upper bound (lub) of B if it is an upper bound of B and if $y' R y''$ $\forall$ other upper bounds y'' of B.

Example:

1. Let D_n be the set of all +ve divisors of 'n', where 'n' is a +ve integer, then $\langle D_n, / \rangle$ is a lattice with the relation '/'.

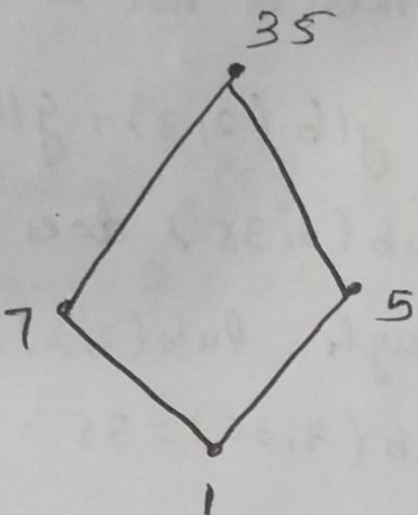
For $n = 35$, we have $D_{35} = \{1, 5, 7, 35\}$, '/'.

$7 \oplus 5$ (lcm(7,5)) = 35.

$7 * 5$ (gcd(7,5)) = 1

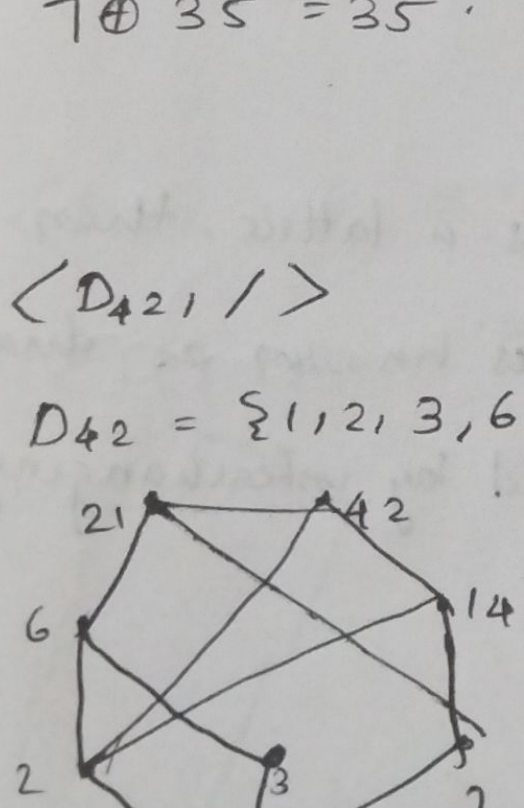
$7 * 35 = 1$

$7 \oplus 35 = 35$.



2. $\langle D_{42}, / \rangle$

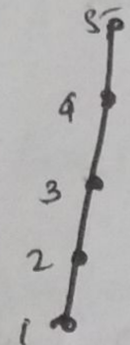
$D_{42} = \{1, 2, 3, 6, 7, 14, 21, 42\}$.



$6 * 14 = 2$.

$6 \oplus 14 = 42$.

3. $A = \{1, 2, 3, 4, 5\}, \leq$



$2 \oplus 4 = 5$

$2 * 4 = 1$

4. Check whether ~~the~~ $A = \{2, 3, 5, 6, 7, 11, 12, 35, 385\}$ under the relation R s.t. xRy if $x|y$ is a lattice?

Ans: This is not a lattice

$\because$ $glb(2,3)$, $glb(5,7)$, $glb(11,35)$, $glb(3,35)$, $lub(3,35)$ does not exist in A.

Though $lub(2,3) = 6$, $lub(5,7) = 35$, $lub(11,35) = 385$.

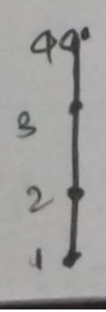
$lub(7,35) = 35$.

Dual or <u>Reversed Lattice</u>

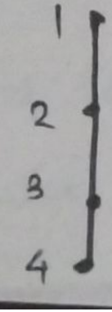
If $(A, \leq) = (A, \vee, \wedge) = (A, \oplus, *)$ is a lattice, then $(A, \geq) = (A, \wedge, \vee) = (A, *, \oplus)$ is known as dual or reversed lattice which is obtained by interchanging glb and lub.

eg:

1. $A = \{1, 2, 3, 4\}$ $\leq \rightarrow R$.



Dual $(A, \geq)$



$lub = 1$

$glb = 4$

Principle of Duality

Any statement of lattices involving the operation $\cdot(*)$ and $+,(\oplus)$ and the relation $\leq$ and $\geq$ remain true if $\cdot(*)$ is replaced by $+(\oplus)$, $+(\oplus)$ by $\cdot(*)$ $\leq$ by $\geq$ and $\geq$ by $\leq$.

Eg: Write the dual of each of the statements.

a) $(a \wedge b) \vee c = (b \vee c) \wedge (c \vee a)$

dual $(a \vee b) \wedge c = (b \wedge c) \vee (c \wedge a)$

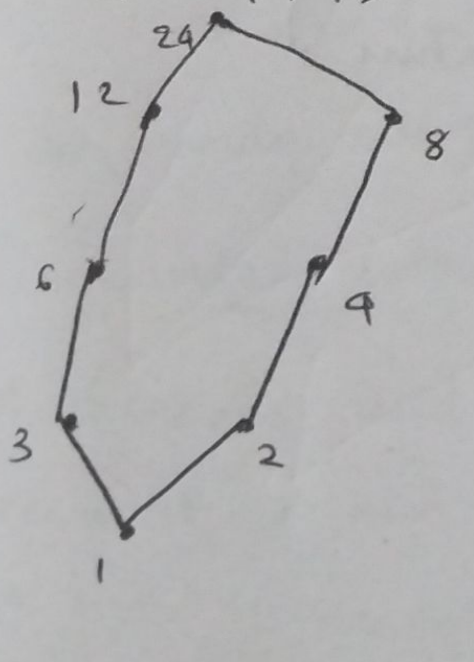
Sub lattice

Let (A, R) be a lattice. A non-empty subset B of A is called a sublattice of A if for any $a, b \in B$, $a \vee b$ and $a \wedge b \in B$.

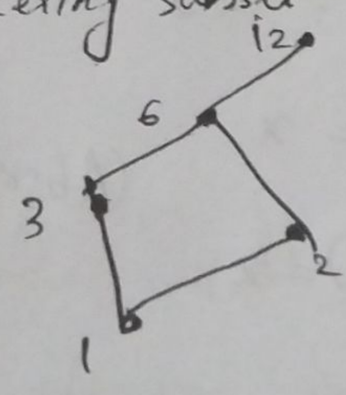
Eg:

1. $\langle D_n, / \rangle$ is a sublattice of $\langle Z^+, / \rangle$

2. $\langle D_{24}, / \rangle$, $D_{24} = \{1, 2, 3, 4, 6, 8, 12, 24\}$.



Considering subset $B = \langle \{1, 2, 3, 6, 12\}, / \rangle$



Properties of lub and glb.

1. $glb(x,y) \leq x$.
   $glb(x,y) \leq y$.

2. $x \leq lub(x,y)$
   $y \leq lub(x,y)$.

3. If $m \leq x$ and $m \leq y \Rightarrow m \leq glb(x,y)$
   $m \leq x\,y$.

4. $x \leq u$ and $y \leq u \Rightarrow lub(x,y) \leq u$. , $x+y \leq u$

Properties of lattice

Let $(A, \leq) = (A, R) = (A, \cdot, +)$. be a lattice, for any $x, y \in A$

1. a) $x + x = x$ $lub(x,x) = x$. (idempotent)
   b. $x \cdot x = x$ $glb(x,x) = x$..

2. a) $x+y = y+x$.
   c) $x \cdot y = y \cdot x$ (Commutative law)

3. a) $x+(y+z) = (x+y)+z$.
   b) $x(yz) = (xy)z$ (Associative)

4. a) $x+(xy) = x$.
   b) $x(x+y) = x$ (Absorption)

Complete lattice

A lattice is called complete, if each of its non-empty subsets have lub ~~aag~~ and glb.

- The greatest element of a lattice, if it exist is denoted by 1 and is called the unit element.
- Similarly the least element, if it exist, is denoted by 0 and is called the zero element.

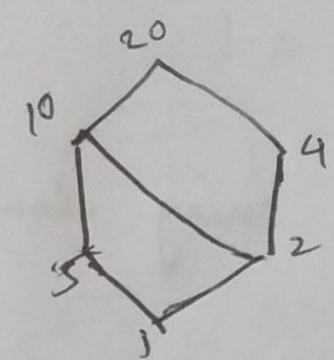
Note:

Every finite lattice are complete

Eg:

1. $(D_{20}, /)$

$D_{20} = \{1, 2, 4, 5, 10, 20\}$.



$\langle Z^+, \leq \rangle$ is not a complete lattice.

$\because$ 1 as the lower bound.

~~Set~~ consider set of all even +ve integers, there is no ~~to~~ lub.

bounded lattice

A lattice is said to be bounded, if it has a greatest element 1 and ~~a~~ least element 0. The elements 0 and 1 are called the bounds of lattice.

Eg: $(P(A), \subseteq)$

Note: Every finite lattice $(A, \leq)$ is bounded.

2. If L is a complete lattice, then it is bounded.

3. Not all bounded lattice are complete lattice.

Complement of an element

for a bounded lattice, an element 'b' is said to be the complement of an element 'a'

if $a \cdot b = 0$ and $a + b = 1$

or

$a \wedge b = 0$ and $a \vee b = 1$

Complemented lattice.

A lattice is said to be complemented if every element has at least one complement

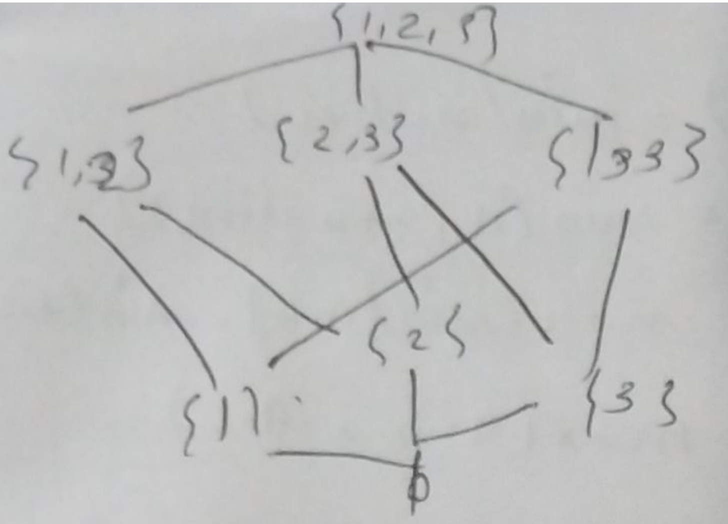
Eg: $\langle P(A), \subseteq \rangle$, $lub = 1 = A$.

$glb = 0 = \phi$.

Let $B \subseteq P(A) \Rightarrow B \subseteq A$.

$B' = A - B$.

Eg: $A = \{1, 2, 3\}$ $\langle P(A), \subseteq \rangle$ be



$\{1,3\}' = \{2\}$.

$\{1,3\} + \{2\} = A = \{1,2,3\} = 1$

$\{1,3\} \cdot \{2\} = \phi$.

$\{1\}' = \{2,3\}$

Distributive lattice

A Lattice $[A, +, \cdot]$ is said to be distributive lattice if for any $a, b, c \in A$

$a + (b \cdot c) = (a+b) \cdot (a+c)$, + is distributive under $\cdot$

$a \cdot (b+c) = a \cdot b + a \cdot c$

Eg: S.T $(P(A), \cup, \cap)$ is distributive.

Ans: for any B, C, D subset of $P(A)$

we have $B \cap (C \cup D) = (B \cap C) \cup (B \cap D)$ and

$B \cup (C \cap D) = (B \cup C) \cap (B \cup D)$.

2. S.T lattice $(Z^+, \leq)$ is distributive.

Ans: Here $lub(a,b) = max(a,b)$

$glb(a,b) = min(a,b)$

Let $a, b, c \in Z^+$, then $a \cdot (b+c) = \min(a, b+c)$

$= \min(a, \max(b, c))$

$= \max(\min(a, b), \min(a, c))$

$= \max(a \cdot b, a \cdot c)$

$= ab + ac$.

III'ly

$a + (b \cdot c) = \max(a, \min(b, c))$

$= \min(\max(a, b), \max(a, c))$

$= \min(a+b, a+c)$

$= (a+b) \cdot (a+c)$.